Cost-Benefit Analysis of New Night Train Service Concepts

Project Report for the Course of Transportation Economics
At the Technical University of Munich for the Winter Term 2023/2024

Lecturer	Prof. Dr. Antoniou Constantinos Chair of Transportation Systems Engineering
Lecturer Assistant	M.Sc. Mohamed Abouelela
Submitted from	Samuel Juhasz-Aba Marsa Tyarpratama Shyam Twanabasu Kiran Chataut
Submitted on	08.03.2024

Contents

List of figures

List of tables

1. Introduction

The concept of night trains allows passengers to travel long distances overnight while benefiting from a higher level of comfort than regular day services (UIC et al., 2013). Passengers may board the train in the evening and arrive, after a longer non-stop journey, at their destination in the morning (UIC et al., 2013). The idea of a night train originated at the "Compagnie Internationale des Chemins des Wagons-Lits" (CIWL), in which beds have been included on a train for the first time (Schiefelbusch, 2016). Today, different travel classes have different amounts of bed per cabin and provide different levels of amenities or on-board service (ÖBB, n.d.).

Until 2016, the German state railway Deutsche Bahn AG (DB) operated own night train services under the name of City Night Line (Hirsch, 2014). However, after a declining demand due to higher competitiveness by high-speed rail and low-cost airlines and a decline in profit, DB gave up their night train business (Hirsch, 2014; UIC et al., 2013). This is when the Austrian state railway ÖBB took over selected routes and rolling stock, which led to a growing number of services and passengers every year, thus their business became profitable (Wüpper, 2023).

Today, a growing number of both private and state-owned night train operators can be observed (Allianz pro Schiene, n.d.). As a private operator, European Sleeper started night train services between Berlin and Brussels or Amsterdam. Moreover, the private Swedish railway company Snälltåget started competing with the state-owned company SJ with adjacent night train services on the exact same route, between Berlin and Stockholm (Allianz pro Schiene, n.d.). Thus, both demand and night train services have been on a rise in recent years, seeing new both private and state-owned companies on the rail market (Allianz pro Schiene, n.d.).

However, similarly to daytime traffic, the competition of passenger rail transport is little, for which different reasons can be identified (Jakobi, 2022). Private companies do not only complain about high track access charges (TAC), lack of subsidies in comparison to state-owned companies, but also have the difficulty to compete against the established network effect of those companies (Jakobi, 2022). Moreover, uniform service concepts exist, lacking in innovation, as they are also often referred to as the "classical night train" in research (UIC et al., 2013). Therefore, this report aims to explore whether new night train concepts are economically feasible, and which of the proposed ones has the highest feasibility.

2. Project Description

2.1 Route selection

The proposed service concepts will be running between Munich to Hamburg via Berlin. The choice has been made since it connects the 3 largest German cities and covers a route in which no regular night train service is provided. Moreover, there is a strong increase of demand and supply during daytime, which might be another opportunity for travelers who do or cannot travel during the day (Deutsche Bahn AG, 2023). Furthermore, an intra-German route strongly facilitates licensing processes and does not require additional costly multi-system rolling stock (Kolvenbach, 2022).

2.2 Identified gaps and formulation of the service concepts

Currently, night trains encounter a multitude of issues, thus possible reasons for operators to avoid this market. First, there is a lack of low-cost carriers. Night trains can be perceived as costly, thus pushing passengers to other modes such as night buses (SRF, 2023). Moreover, long travel times and outdated rolling stocks might attract passengers more towards modern high-speed rail by day (Ehrbar, 2023). Another common issue with night trains is that they are often not used during daytime, thus staying in depots (Imwinkelried, 2023). Their single occupancy of a bed during a trip is also a challenge if the proposed service should be profitable (Imwinkelried, 2023). Based on those stated problems, the following service concepts have been elaborated:

- **Service Concept 1:**

A low-cost service that provides almost no on-board amenities. In other words, passengers would be required to bring their bed sheets or sleeping bags/supplies. This service would run with 4 6-bunk beds couchette cars and 2 seating cars. With a travel time of about 8h15min., it would stop in Munich and Augsburg, and run non-stop to Berlin before reaching its destination Hamburg.

- **Service Concept 2:**

To tackle the issue of slow night trains, a high-speed service will be proposed. With updated rolling stock, based on the newest generation of NightJet trainsets, a travel time of 6 hours and 30 mins. could be reached. It would run with one 2-bed sleeper coach, 3 6-bunk beds couchette cars and 2 seating cars and stop at the same stations as the low-cost service.

- **Service Concept 3:**

Lastly, to avoid high depot cost and additional potential profit generation at daytime, the third service concept would use regular rolling stock with a 2-bed sleeper coach, 3 6-bunk beds couchette cars and 2 seating cars. Its travel time would be 8 hours and 15 mins one way. By night, it would stop at the same stations as service concept 1 and 2, whereas by day, it would additionally stop in Würzburg, Fulda, Bad Hersfeld, Eisenach, Gotha, Erfurt, Halle and Bitterfeld, thus running a similar service pattern as adjacent InterCity or Flix Train services.

3. Methodology

3.1 Introduction and Overview of the CBA Methodology

A systematic and quantitative Cost-Benefit Analysis (CBA) is conducted to evaluate the economic feasibility of new potential night train service concepts. This approach allows us to assess the service's economic, social, and environmental impacts, thus ensuring a comprehensive understanding of its benefits and costs over the period of analysis. Analysis has shown the actual costs and benefits generated from the project by considering both direct and indirect impacts on the economy, environment, and society. The monetary values used to estimate costs and benefits are derived from extensive literature research. Furthermore, by evaluating three distinct night service concepts, the aim is to enhance the likelihood of selecting the service concept that offers the greatest overall benefit.

The first step involves a thorough estimation of all the project's costs and monetarized benefits over its expected lifetime. Then, to assess the financial success of introducing new night train services, it is essential to look at various key parameters. Following parameters are crucial in providing a detailed understanding of the project's economic and financial outcomes:

- **Project Duration and Economic Growth Rate**

The project's time horizon is the year 2040, so a period of about 16 years will be assessed. Therefore, all cash-flow forecasts and financial predictions have covered this period to match the project's potential lifetime.

- **Discount Rate**

Moreover, to calculate the present value of the future cash flows, a discount rate of 3.5 % is used. The discount rate used in the analysis is based on the guide of Cost-Benefit analysis by the Directorate General Regional Policy of the European Commission (Guide to Cost-Benefit Analysis of Investment Projects, n.d.).

- **Calculation of the Net Present Value (NPV)**

Hence, the Net Present Value (NPV) can be calculated with the mentioned parameters, which is the difference between the discounted total social benefits and costs. A positive Net Present Value signifies an economically feasible project, in which the benefits outweigh the cost of the project (Guide to Cost-Benefit Analysis of Investment Projects, n.d.). The NPV can be mathematically expressed as followed:

$$\text{ENPV} = \sum_{t=0}^{t=n} \frac{(B_t - C_t)}{(1+r)^t}$$

ENPV = Economic Net Present Value
B_t = Benefits (income) in year t

C_t = Costs (expenses) in year t
r = Discount rate

- **Calculation of the Benefit-Cost Ratio (BCR)**

The Benefit-Cost Ratio (BCR) offers a different approach to assess the present value of potential alternatives by comparing the benefits against the costs. Unlike the approach of determining the Net Present Value (ENPV) through the subtraction of the present value of costs from the present value of benefits, the BCR methodology involves the division of the present value of benefits by the present value of costs (Guide to Cost-Benefit Analysis of Investment Projects, n.d.). Similar to the NPV, a BCR that is greater than 1 discloses the economic feasibility of a proposed project. The BCR can be mathematically expressed as followed:

$$BCR = \frac{\sum_{t=0}^{t=n} \frac{B_t}{(1+r)^t}}{\sum_{t=0}^{t=n} \frac{C_t}{(1+r)^t}}$$

BCR = Economic Net Present Value
B_t = Benefits (income) in year t
C_t = Costs (expenses) in year t
r = Discount rate

- **Sensitivity analysis**

Uncertainty is present in all economic evaluations, and it relates to several components of the analysis. An analyst often relies on estimates for variables that can't be predicted with 100% certainty when conducting a cost-benefit evaluation. As a result, the result of the CBA is subject to risk and uncertainty, as it cannot be determined with complete confidence (Bonner, 2022). Sensitivity analysis is a straightforward way to address uncertainty in cost-benefit analysis; it involves the evaluation of which outcomes are sensitive to the assumed inputs in the cost-benefit analysis. A sensitivity analysis aims to determine how much the net social benefit changes when assumed input values change. Most importantly, it identifies how independent inputs into the policy, program, or project can affect the net present value – and, consequently, the potential profitability of the project. Any cost-benefit analysis should be subject to a sensitivity analysis, making it the last major step of the CBA before providing the recommendation or action (Bonner, 2022).

According to Briggs et al. (1994), there are four types of uncertainty, including dealing with data inputs where no clear sample exists, attempting to increase the generalizability of the study, handling uncertainty associated with attempts to extrapolate away from the primary data source to make the results more comprehensive and to explore the implications of selecting a particular analytical method from amongst alternatives when no widely accepted approach exists. Table 1 shows that only simple sensitivity analysis–the most common form, where one or more components of an evaluation are varied across a plausible range–can deal with all four forms of uncertainty, which we used for the last major step of the CBA in our night train study, namely the one-way simple sensitivity analysis.

Table 1. Appropriateness of each of sensitivity analysis in dealing with the various types of uncertainty (Briggs et al., 1994)

Sensitivity analysis	Uncertainty: Data variability	Generalisability	Extrapolation	Analytical method
Simple sensitivity analysis	✓	✓	✓	✓
Threshold analysis	✓	✓	✓	×
Analysis of extremes	✓	○	✓	×
Probabilistic sensitivity analysis	✓	○	○	×

✓ = Generally useful
○ = Potentially useful in certain contexts
× = Unlikely to be useful

3.2 Demand Estimation for Night Train Services

The current demand estimation for night train services is limited by a lack of quantitative secondary data. Therefore, values based on the general German modal split and newspaper indications will be used to estimate a potential night train demand. As there is no precise and available data for the intercity travel demand with modes other than air traffic, the travel demand by air between major cities along our night train route has been used. The following values of air travel demand between the major cities on our proposed routes are as follows:

Table 2. Air travel demand (Oui au train de nuit., 2023)

Route	Passengers per year
Hamburg- Munich	1.739.978
Berlin-Munich	1.934.021
Total passengers (Munich – Hamburg & Berlin)	3,67 million

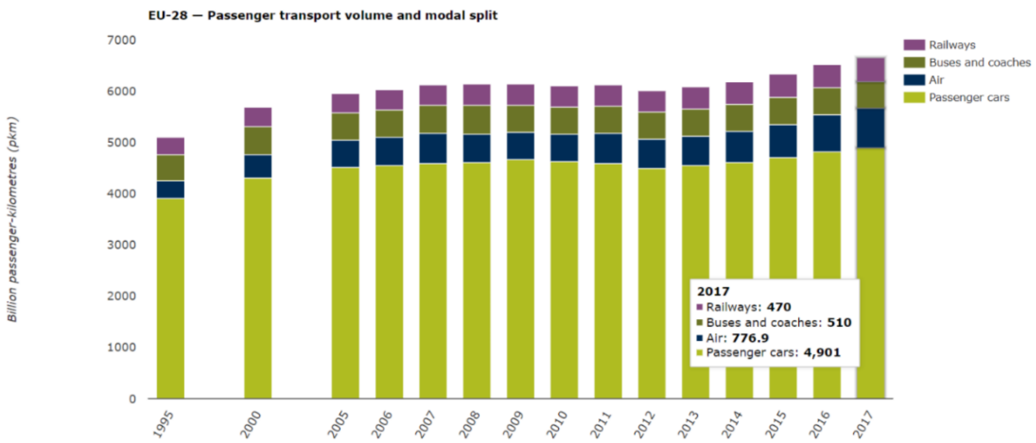


Figure 1. Passenger transport volume and modal split for long distance travel (European Environment Agency)

With the help of the bar chart above, the modal split of railways for long distance passenger travel compared to air transport can be identified. Therefore, following calculations have been done:

$$Air\ Modal\ Split\ (EU) \quad = \frac{776{,}9}{470 + 510 + 776{,}9 + 4901} * 100 = 11{,}67\ \%$$

$$Railway\ Modal\ Split\ (EU) \ = \frac{470}{470 + 510 + 776{,}9 + 4901} 470470 + 510 + 776{,}9 + 4901 * 100 = 7{,}05\ \%$$

$$Total\ Amount\ of\ Passengers\ (Munich - Hamburg - Berlin)$$

$$= \frac{3{,}67\ million\ passengers\ per\ year}{11{,}67\ \%} = 31{,}45\ million\ passengers\ per\ year$$

$$Total\ Amount\ of\ Railway\ Passengers\ (Munich - Hamburg - Berlin)$$
$$= 7{,}05\% * 31{,}45\ million\ passengers = 2{,}21\ million\ passengers\ per\ year$$

For 2014, the Deutsche Bahn AG Economic Committee reports 2,7 million passengers per year on night train services. Media reports estimated a demand in 2015 of around 1,3 to 1,5 million travelers in night train traffic, although these figures exclude passengers in seated carriages. At the end of 2016, "Bahn für Alle" estimated demand for night trains at 3 million travelers (P.P.T.V. AG et al., 2017). For our further calculations, the number of night train passengers in Germany will be set to 3 million passengers per year. Moreover, according to Deutsche Bahn AG's report in 2020, the number of long-distance rail passengers was estimated at about 139 million passengers per year. Therefore, the following calculations have been made:

$$Share\ of\ Nigh\ Train\ Passengers\ in\ Germany$$

$$= \frac{3}{139} * 100 = 2{,}16\ \%$$

$$Night\ Train\ Passengers\ on\ the\ Route\ Munich - Hamburg - Berlin$$

$$= 2{,}16\% * 2{,}21\ million\ passengers\ per\ year = 48.000\ passengers\ per\ year$$

The following flow chart summarizes the methodology that has been described above:

Moreover, DB AG announced in October 2016 the following increases in travel demand and number of services provided compared to the same period in 2015 (P.P.T.V. AG et al., 2017):

+ 14,4% in the transport performance of all night trains

+ 20,2% demand in the Hamburg/Amsterdam-Frankfurt-Munich/Zurich network

+ 5,9% in all sleeping and couchette cars (with slightly reduced space)

+ 27,2% in all seated carriages (with significantly increased space)

Therefore, the night train demand is further increasing.

According to ÖBB, the trend of night trains is on the rise, as they plan to operate 33 new NightJet trainsets from autumn 2023 and onwards. In 2022, they recorded a ridership of 1,5 million passengers per year, hence new routes are expected to increase demand (Europe Zig Zag, 2023).

Indeed, the current night trains fall short of fully utilizing the line's potential. Some existing night trains suffer from limited capacity, outdated facilities, and inadequate marketing efforts. However, there remains significant room for enhancement, even on established routes. For instance, high-demand routes could offer additional services during the same night. Notably, night trains serve as a comfortable land-based alternative for air travelers to consider (Oui au train de nuit., 2023).

3.3 Costs

In the first step of a CBA, all costs until the project's completion and its entry into operation must be calculated (Naess, 2006). Direct expenses include both capital and operational costs, which are essential for establishing and operating the proposed services. On the other hand, indirect expenses capture all external costs, particularly those related to environmental impacts. The methodology used to quantify all costs has been elaborated in the following section:

3.3.1 Capital Costs

First, capital costs will be enumerated, which are substantial investments in the physical assets required to start a new night train service. These costs are a one-time investment in physical assets incurred at the start of a project.

- **Rolling Stock & Locomotives Cost**

Locomotives are the powerhouse of railways, and rolling stocks are passenger coaches equipped with various amenities to ensure passenger comfort (Dailydka, 2010). Night trains are commonly equipped with couchette cars, sleeping cars, and seated and reclining cars to accommodate overnight travel (European Parliament et al., 2017).

The capital cost for the high-speed train set is estimated based on the price of ÖBB's new NightJet model (Maier, 19 November 2023), which is supposedly around 20 million euros. Similarly, costs for low-cost and day-use trains, totaling around 7 million, are calculated based on the price per unit of sleeping cars, couchette cars, seated and reclining cars, and locomotives. The unit values are taken from BVWP 2030, and the assumed lifespan for the rolling stock has been set at 15 years (The 2030 Federal Transport Infrastructure Plan, n.d.). Moreover, a linear depreciation across the operational life of the rolling stock has been considered and included in the price per unit of rolling stocks.

3.3.2 Operational costs

After the capital costs have been enumerated, operational costs will be elaborated. They comprise all the recurring expenses associated with the day-to-day running of the project once it is operational. These are costs incurred regularly and necessary for the ongoing functions of the night train service.

- **Personnel Cost**

Personnel costs include the salaries of the train driver, head of the train attendant, and train attendant. For our project we have used the valuation of personnel costs as outlined in the BVWP (Bundesverkehrswegeplan) 2016 and projected for 2030 (Fraunhofer et al., 2017). Extra remuneration for night, Sunday, and holiday work at 15% and a similar overhead charge for administrative costs are included in these rates. This aspect considers the complex scheduling requirements for train drivers and attendants, such as standby times, vehicle setup and dismantling times.

- **Energy Cost**

The expenses associated with consuming electrical power for the operation of trains are calculated according to the approach provided by BVWP (Bundesverkehrswegeplan) 2016. It considers the direct traction energy required to move the trains at various speeds and the additional energy onboard systems consume (Fraunhofer et al., 2019). Energy consumption is measured in kilowatt-hours per ton kilometer (kWh/km) to account for the weight of the passengers and wagons for each night train service. The resulting value in kWh per journey is multiplied by Germany's energy prices, as detailed in the BVWP (Bundesverkehrswegeplan) 2016.

- **Route Price**

Route costs are a critical component in the economic feasibility analysis of rail services. These prices reflect the costs per route kilometer that rail operators are expected to pay for using the rail infrastructure in each country. The route prices used in our calculation are obtained from route prices per kilometer in Germany based on data from the CIS Online Tool. The price is adjusted to the new route pricing system, effective from 10 December 2017 (Fraunhofer ISI., 2017).

- **Station Fees**

Station fees also play a significant role in the operation of our passenger train service and the costs of using the routes. These fees differ widely based on the amenities and infrastructure available at each station (Fraunhofer ISI., 2017). The price per station is taken from the DB Netz AG's TPS (Trassenpreissystem) Track Access and Station charges catalog (Deutsche Bahn AG, 2022). Depot costs for Service 1 and Service 2 are also included in this station's fees and have been gathered from DB Netz AG's APS Price catalog (Anlagenpreissystem).

- **Track Access Charge**

Track Access Charges (TAC) are fees railway operators pay to access and use the rail infrastructure (Bundesnetzagentur - Charges, n.d.). New findings from Back-On-Track Europe, focusing on Germany, illustrate that lowering track access charges (TAC) for night trains presents a strategic approach to revive the market (e. V, 2023). The price reference for the calculation is taken from the same Back-On-Track report (e. V, 2023).

- **Traffic Management System**

Expenses for the safety and efficiency of night trains, such as signaling systems and software for train routing and scheduling, are also considered in the calculations. The annual prices are approximated based on the operational cost of existing night train services.

- **Insurance Cost**

Insurance expenses for train services cover payments made to mitigate risks like accidents, damage to trains, liability for passenger injuries, and other unexpected incidents. Despite the absence of research papers directly focusing on the yearly insurance expenditures for trains, annual cost estimates are derived by comparing external costs for each night train service.

- **Maintenance, Inspection, and cleaning costs**

The International Union of Railways (UIC) and the comprehensive study conducted by Garcia and Espanoles (2010) have provided a deeper insight into the multifaceted nature of maintenance, inspection, and car cleaning expenses. These maintenance expenses comprise fixed costs, including management and technical support, and variable costs, which account for the wear and tear associated with train operation. Inspection costs include expenses associated with inspecting and maintaining the train at workshop facilities (Fraunhofer ISI et al., 2017). The maintenance expense per kilometer for each night train car and inspection cost per kilometer and car are obtained from the abovementioned study. Similarly, the cleaning costs are calculated per car and trip depending on the car type, and the service class is also calculated.

3.3.3 External costs

The environmental cost of new night train services cannot be neglected. The operation of services produces environmental costs regarding accidents, air pollution, well-to-tank, and noise. Factors like potential risk and consequences of accidents occurring during night-time operations, energy production, and supply chain before usage by the trains, and the impact of noise from night trains on surrounding areas are considered, and costs are internalized. Even though it is negligible compared to other modes, air pollution cost is also considered in the calculation. Eurocent per passenger kilometer (€-cent/km) is calculated from the "EU Handbook on external cost of transport" (CE Delft et al., 2020). The balance between average and marginal costs is maintained in the calculations.

3.4 Benefits

In the following section, both direct and indirect benefits will be quantified.

3.4.1 Direct Benefits

- **Revenue**

The ticket prices are the main sources of revenue for the night train. The ticket price was calculated so that it covers the operational costs of the service. The formula used and the calculated values can be found in the table below.

$$Ticket\,\mathrm{Pr}\,i\,ce = \frac{Total\cos t\ p.a.}{Passenger\ capacity}$$

Table 2. Ticket price calculation

	Ticket price [€]	**Ticket price in 2040, including a yearly inflation rate of 2,9% [€]**
Service Concept 1	45	65
Service Concept 2	90	130
Service Concept 3	50	75

The ticket prices are susceptible to change with market dynamics. In our calculations, we have calculated the average ticket prices by adding the total annual operational cost for each service type and dividing it by the total passenger capacity of the service. In the case of an actual market implementation dynamic ticket pricing could be implemented, which is beyond the scope of this cost-benefit analysis.

The total revenue generated from the ticket sales is the average ticket price multiplied by the average occupancy rate and the total train capacity.

According to research led by Bird et al. (2019), City Night Line night trains had an average occupancy of 45-50% in 2016. To achieve financial sustainability, they would need to increase occupancy to 60-65%. Meanwhile, ÖBB's NightJet services experience seasonal demand, with load factors ranging from 70% in summer to less than 50% after Christmas. Routes to major German business centers such as Munich and Hamburg show less seasonality. As per DB, the total number of passengers carried at night is expected to grow, but the demand will increasingly be for low-fare overnight trains using day stock instead of high-quality night trains using specialized rolling stock. (Bird et al., 2019).

Therefore, we are using an average occupancy of 65 % in this cost-benefit analysis. As governments are promoting sustainable modes of transport in combination with tax changes and reduction of short haul domestic flights, people are choosing environmentally friendly travel patterns and night trains are continuously innovating with improved rolling stock, better connectivity, and enhanced amenities (Allianz pro Schiene, n.d.).

Moreover, the German government does not provide subsidies for long-distance transport services. Instead, it encourages operators of long-distance transportation services to develop self-sufficient business plans. Consequently, we have excluded subsidies as a source of income in our analysis.

3.4.2 Indirect Benefits

- **Day-Time Travel Time Saving**

As the passenger travels during the night-time, they save productive daytime which otherwise would have been spent on travelling. Also, passengers can sleep during the travel which omits an "unproductive" night-time of 8 hours and makes it comparable to airplane travel time. This assumption can be calculated by multiplying the hours saved compared to flight with the economic value of time:

$$Hours\ Saved = Sleep\ Time\ (8hrs) + Flight\ Time\ (1{,}5hrs) - Nighttrain\ Travel\ Time$$

The economic value of time depends on varied factors such as mode of transport, purpose of travel, economic status of the person and more. The average value of time for the passengers in night trains can be set to 26,2 €/h (Quinet, 2013).

- **Travel Time Saving**

Night trains are slower than day trains as they provide enough time for sleep, split during their trip to serve several destinations, and often give priority to freight trains (Bird et al., 2017). However, given that the perception of travel time varies across transport modes and depends on the activity that travelers are doing while traveling, night train travel time might be perceived as shorter given the possibility of sleeping during parts of the trip (Juschten & Hossinger, 2020). This travel time saving can be calculated based on a mode shift. Therefore, it is calculated by multiplying the time saved by choosing night trains instead of other modes of transport (air) with the value of time for passengers.

$$Time\ Saved = Airplane\ Travel\ Time - Night\ Train\ Travel\ Time + Offset\ Time$$

- **Travel Comfort Benefits**

The level of comfort is an important determinant for traveling by night train, and in particular the number of people a compartment is shared with, hence the "privacy" aspect is important (Bulková, Dedík, & Gašparík, 2022). In terms of monetary value, comfort benefit is the benefit of additional comfort (represented by a star rating system, e.g., normal train comfort is rated at 2 stars) multiplied by the value of willingness to pay by passengers for the upgraded comfort. The following parameters have been included in the calculations:

Table 3. Comfort rating parameters

Passenger cars	Rating stars
Sleeping cars (A)	5
Couchette cars (B)	3
Seated and reclining cars (C)	2

Based on the train configuration, we calculated the average star rating of the service concept. Later adjustments were made based on the number of stops i.e., a reduction of comfort rating by 0.07 stars was done per stop.

$$Comfort\ Benefit = (Adjusted\ Star\ Rating - Basic\ Star\ Rating\ per\ Seat) * Willigness\ to\ pay$$

The willingness to pay for each increase in star rating was set to 10 €. As the star rating increases, people are more willing to pay. In other words, higher-star accommodations tend to attract a higher price point.

3.4.3 Intangible benefits

- **Environmental benefit**

Environmental benefit can be understood as the benefit of using sustainable night trains compared to other modes. Air travel is notorious for its high carbon emissions. Thus, by encouraging passengers to shift from planes to night trains, we can calculate the reduction of emissions. This reduction of air pollution represents the environmental gain that would have otherwise been incurred if those passengers had chosen air travel.

Table 4. Air to Train Passenger Shift

Total airway passenger (Munich- Hamburg & Munich-Berlin route)	1 739 978+ 1 934 021 **=3,67** million passengers
Strong willingness of mode shift to railways (two way)	14 % of air way travellers (Donat, 2021) =**0,5** million
Percentage of night passengers	**2,16%**
Passengers expected to switch to night train one way (Munich – Hamburg & Munich-Berlin route)	1/2 of 2,16% of 0,5 million = approx. **6000** passengers

Environmental benefits are calculated by the difference between the environmental cost of travel by air and by train. The environmental costs of both modes are calculated as follows:

$$Environment\ cost\ due\ to\ travel\ by\ air = Number\ of\ passengers\ *\ total\ travel\ distance\ *\ specific\ emission\ cost\ per\ passenger\ km\ for\ airways$$ and

$$Environment\ cost\ due\ to\ travel\ by\ train = Number\ of\ passengers\ *\ total\ travel\ distance\ *\ specific\ emission\ cost\ per\ passenger\ km\ for\ train$$

According to the European Commission's handbook from 2019, the specific emission cost is 0.3 € cents per passenger kilometre for air travel, while for electric trains, it is significantly lower at 0.01 € cents per passenger kilometre.

- **Social Benefits**

One advantage of the night train is precisely its ability to serve many medium-sized towns in the evening and morning, thus offering a much more refined service than air travel. Night trains can thus address the lack of accessibility by facilitating travel for people living in regions that are poorly connected today. On line sections with poor accessibility today, there is therefore a potential to induce traffic by the introduction of night trains. However, this is difficult to estimate. (Oui au train de nuit., 2023)

4. Results

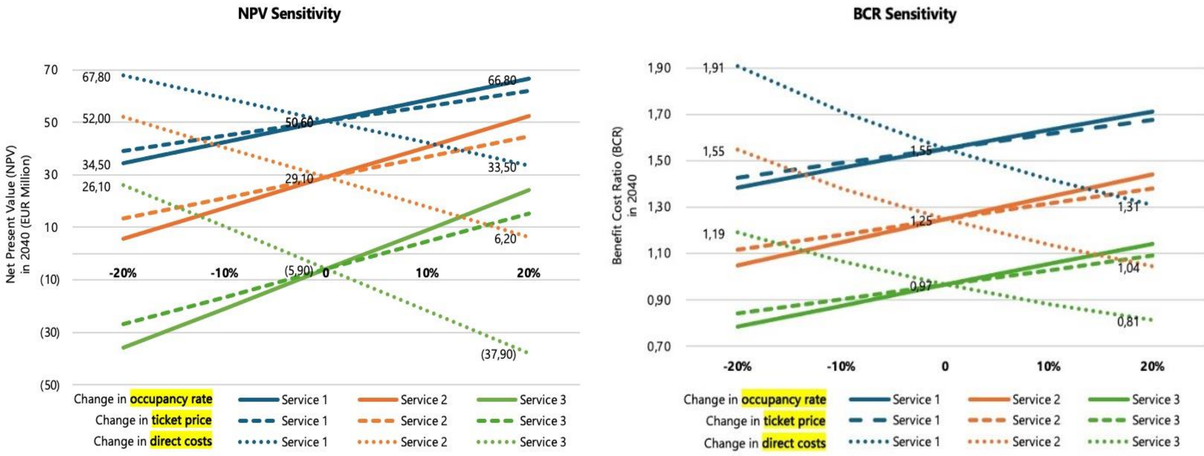
Table 5. CBA results

	Service 1	Service 2	Service 3
Costs	**Initial Investment Cost [thousand €]**		
	7.000,00	20.020,00	7.000,00
	Operational Costs [thousand €/annum]		
	6.004,00	7.224,00	11.678,00
	External Costs [thousand €/annum]		
	475,00	269,00	680,00
Σ Cost	117.062,00	147.266,00	216.949,00
Σ Cost (incl. 3.5% discounting)	91.774,00	118.044,00	168.734,00
Benefits [thousand €/annum]	**Direct Benefits**		
	4.355,00	5.993,00	8.043,00
	Indirect benefits		
	6.387,00	5.164,00	4.291,00
	Intangible Benefits		
	133,00	75,00	105,00
Σ Benefits	184.832,00	190.912,00	211.409,00
Σ Benefits (incl. 3.5% discounting)	142.365,00	147.069,00	162.859,00
NPV [thousand €]	**Net-Present-Value (incl. 3.5 % discounting)**		
	50.591,00	29.025,00	-5.875,00
BCR	**Benefit-Cost Ratio (incl. 3.5 % discounting)**		
	1,55	1,25	0,97

Service Concept 1 (SC1), which offers basic amenities, stands out as the most feasible project with a Benefit-Cost Ratio (BCR) of 1,55. Despite having the lowest annual cost, the low-cost variant of SC1 still provides comparable benefits. Service Concept 2 (SC2), involving a high-speed night train service, follows closely with a BCR of 1,25. Whereas, service Concept 3 (SC3), combining regular night trains and day trains, has a BCR of 0,97, rendering it economically unfeasible. The Net Present Value (NPV) analysis aligns with these findings.

In summary, SC1 emerges as the most viable option, while SC3 is economically infeasible. The results in Table 5 also suggests that the service concepts SC1 and SC2 is a profitable business model.

5. Sensitivity Analysis



(a) (b)

Figure 2. Sensitivity to change in three different components on all service concepts (a) NPV; (b) BCR (Authors)

In this study, we evaluate the net present value (NPV) and the benefit-cost ratio (BCR) sensitivity to three various components of our economic evaluation. We analyze it based on ± 20% changes in occupancy rates of 65%, the ticket price, and direct costs of our basic calculation. Figure 1a shows that among the components analyzed, a change in direct costs has the most significant influence, followed closely by a change in occupancy rate. For example, a 20% change in direct costs will result in a 33% or equal to € 17,2 million difference from the base calculation for the service concept 1, a 78% or equal to € 22,9 million, and a 546% or equal € 32 million difference for the concept 2 and 3 respectively. This also explains that the service concept 3–night train service also running in the daytime–is the most sensitive to change because it involves twice the number of services compared to the other services. Furthermore, if the average occupancy rate of service concept 1 drops to 45%, the net present value will decrease by 32% from € 50,6 million to € 33,5 million in 2040. On the contrary, if the rate increases to 85%, it will increase by 32% from € 50,6 million to € 67,80 million in the same year. From then on, the same pattern will apply to the other service concepts.

On the other hand, figure 2b depicts the BCR sensitivity to change in the same components measured in NPV sensitivity. It fundamentally shows a similar pattern across all services, where changes in direct costs also have the most significant impact on the benefit-cost ratio than other components. However, it is interesting to see that the margin of the BCR between changes in direct costs and occupancy rate is significantly higher than in NPV sensitivity. Moreover, it appears as a non-linear curve, which indicates that small changes in the direct costs–decrease, in this case–may lead to a higher magnitude of increase in the benefit-cost ratio. For instance, if direct cost decreases by 20%, the benefit-cost ratio will increase by 23% from 1,55 to 1,91, whereas if the direct cost increases by 20%, the benefit-cost ratio will decrease by only 15% from 1,55 to 1,31.

6. Conclusion

In summary, the comprehensive cost-benefit analysis conducted in this study illuminates the multifaceted aspects of new night train service concepts, offering a nuanced understanding of its financial implications and potential societal benefits. Our analysis shows that service concepts 1 and 2 will be economically feasible in 2040, showing a benefit-cost ratio of 1,55 and 1,25, respectively. On the contrary, service concept 3 will need more time to reach a profit, with a BCR of 0,97 in the same year. Consequently, service concept 1 – a low-cost night train with almost no onboard amenities – emerges as the most profitable option compared to service concept 2 – the high-speed night train – and service concept 3 – additional daytime operation. In addition, based on our sensitivity analysis, it is important to note that we need to pay more attention to the assumptions made in calculating the direct costs due to their significant impact on the net present value and benefit-cost ratio in 2040. In conclusion, service concept 1 is the best option for the new night train service, alongside paying extra attention to direct cost calculation.

Limitations of this study include the uncertainty of the number of costs and benefits to be included in the analysis; inaccurate assumptions on monetizing the externalities; discounting the future benefits and costs can lead to undervaluing of long-term benefits; subjectivity introduces complexity in comparing benefits across different service concepts; and social benefits, i.e. equity are currently unquantified. These limitations might introduce possible improvements in the night train study. Firstly, monetizing equity and including it in benefit calculation. Lastly, implementing multi-way simple sensitivity analysis, sometimes referred to as 'scenario analysis', can be used to explore the implications of alternatives, each of which affects several parameters in an evaluation.

7. References

Allianz pro Schiene. (n.d.). Die Rückkehr der Nachtzüge. Allianz pro Schiene. Retrieved February 25, 2024, from https://www.allianz-pro-schiene.de/themen/personenverkehr/nachtzuege/

Bird, G., Collins, J., Da Settimo, N., Dunmore, D., Ellis, S., Khan, M., … Vollath, C. (2017). Research for TRAN committee - passenger night trains in Europe: The end of the line?. In Policy Department for Structural and Cohesion Policies, European Parliament. https://doi.org/10.2861/414087. https://re.public.polimi.it/handle/ 11311/1021757

Bird, G., Collins, J., Da Settimo, N., Dunmore, D., Ellis, S., Khan, M., ... & Nistri, D. (2018). Research for TRAN Committee-Passenger night trains in Europe: the end of the line?.

Bonner, S. (2022). *Social Cost Benefit Analysis and Economic Evaluation*. The University of Queensland. https://doi.org/10.14264/2c7588c

Briggs, A., Sculpher, M., & Buxton, M. (1994). Uncertainty in the economic evaluation of health care technologies: The role of sensitivity analysis. *Health Economics*, *3*(2), 95–104. https://doi.org/10.1002/hec.4730030206

Bulková, Z., Dedík, M., & Gašparík, J. (2022) The Potential of Night Passenger Trains in Europe in the Post-Pandemic Period. Transport technic and technology, 18(2), 7-14.

Bundesnetzagentur - Charges. (n.d.). Www.bundesnetzagentur.de. Retrieved 24 February 2024, from https://www.bundesnetzagentur.de/EN/Areas/Rail/Companies/Charges/start.html

Curtale, R., Larsson, J., & Nässén, J. (2023). Understanding preferences for night trains and their potential to replace flights in Europe. The case of Sweden. Tourism Management Per-spectives, 47, 101115.

Dailydka, Stasys. (2010). Choosing railway Vehicles for carrying passengers. Transport. 25. 10.3846/transport.2010.02.

Das Trassenpreissystem 2024 Änderungen und Neuerungen.(2024). https://www.dbinfrago.com/web/schienennetz/leistungen/trassen/trassenpreise/rassenpreissystem/tps-2024-10879322

Deutsche Bahn AG. (2023, September 29). Fahrplan 2024: Bis zu 25 Prozent mehr Sitzplätze zwischen München und Berlin. https://www.deutschebahn.com/de/presse/presse-regional/pr-muenchen-de/aktuell/presseinformationen/Fahrplan-2024-bis-zu-25-Prozent-mehr-Sitzplaetze-zwischen-Muenchen-und-Berlin-11893520

Deutsche Bahn AG. (2022). List of station prices 2022 (valid as of 01 January 2022). DB Station&Service AG.https://www1.deutschebahn.com/resource/blob/6539850/164533bea0bfd8d7ca63e754e57151ec/list-of-station-prices-2022-data.pdf

Donat, O. C. S. K. L. (2021, March 29). European public opinion poll shows support for shifting flights to rail. Germanwatch e.V. https://www.germanwatch.org/en/20045

Ehrbar, S. (2023, July 28). Verspätungen, fehlende Betten, dreckige WCs: Krisenfall Nachtzug. Luzerner Zeitung. https://www.luzernerzeitung.ch/wirtschaft/reisen-verspaetungen-fehlende-schlafwagen-dreckige-wcs-die-nachtzuege-aus-der-schweiz-bieten-ungewollte-abenteuer-ld.2492082

European Commission. (2019). Handbook on the external costs of transport. Delft: CE Delft

European Environment Agency. Passenger Transport Volume and Modal Split. Available online: https://www.eea.europa.eu/data-and-maps/daviz/passenger-transport-volume-5

European Parliament, Directorate-General for Internal Policies of the Union, Leach, T., Preti, A., Dunmore, D. (2017). Passenger night trains in Europe: the end of the line? research for TRAN Committee, European Parliament. https://data.europa.eu/doi/10.2861/414087

Europe Zig Zag (2023). Nightjet and the demand for more sleeper trains in 2023. Europe Zig Zag. https://www.europezigzag.com/2023/05/nightjet-and-the-demand-for-more-sleeper-trains-in-2023/

Fraunhofer, I. (2017). Wissenschaftliche Beratung des BMVI zur Mobilitäts- und Kraftstoffstrategie PTV Planung Transport Verkehr AG

Hirsch, A. (2014, November 10). Ende einer Ära: Die Bahn stellt die Schlafwaggons ein. FAZ.NET. https://www.faz.net/aktuell/reise/ende-einer-aera-die-bahn-stellt-die-schlafwaggons-ein-13255813.html

Imwinkelried, D. (2023, September 7). Nightjet-Züge: Wie DB und ÖBB mit fehlender Rentabilität von Nachtzügen hadern. https://www.handelsblatt.com/unternehmen/handel-konsumgueter/nightjet-zuege-wie-db-und-oebb-mit-fehlender-rentabilitaet-von-nachtzuegen-hadern/29331588.html

Jakobi, L. (2022, December 27). Warum es die Konkurrenz der Deutschen Bahn so schwer hat | MDR.DE. MDR. https://www.mdr.de/nachrichten/deutschland/wirtschaft/deutsche-bahn-mitbewerber-konkurrenz-100.html

Juschten, M., & Hossinger, ¨ R. (2020). Out of the city–but how and where? A modedestina-tion choice model for urban–rural tourism trips in Austria. Current Issues in Tourism, 1–17. https://doi.org/10.1080/13683500.2020.1783645

Kolvenbach, M. (2022, November 19). Deutschland bremst Europas Bahnen aus. tagesschau.de. https://www.tagesschau.de/investigativ/swr/bahn-infrastruktur-105.html

Lena Donat, M. T., Janeczko, L., Majewski, A., Lespierre, T., Fosse, J., Vidal, M., & Gilliam, L. (2021). Hop on the train: A rail renaissance for Europe how the 2021 European year of rail can support the European green Deal and a sustainable recovery. https://germanwatch.org/en/19680.

Maier, J. (2023, November 19). Finally on Track: ÖBB's Next Generation Nightjet- Back-on-Track. https://back-on-track.eu/finally-on-track-obbs-next-generation-nightjet/

Naess, P. (2006). Cost-benefit analyses of transportation investments: Neither critical nor realistic. Journal of Critical Realism. 5. 32–60. 10.1558/jocr.v5i1.32.

ÖBB. (n.d.). Nightjet. Nightjet. Retrieved February 25, 2024, from https://www.nightjet.com/de/

Oui au train de nuit. (2023, March 24). *1 passager aérien sur 2 pourra bénéficier des trains de nuit, si l'Europe investit*. Mediapart. https://blogs.mediapart.fr/ouiautraindenuit/blog/290821/1-passager-aerien-sur-2-pourra-beneficier-des-trains-de-nuit-si-l-europe-investit

P. P. T. V. AG, Walther, C., Schäfer, T., Karthaus, D., & Schade, W. Wissenschaftliche Beratung des BMVI zur Mobilitäts-und Kraftstoffstrategie.

Quinet, E. (2013). L'évaluation socio-économique en période de transition. Rapport du groupe de travail présidé par Émile Quinet, Commissariat général à la stratégie et à la prospective, 1.

Schiefelbusch, M. (2016). Visions of Rail Development–The 'Internationality of Railways' Revisited1. In Linking Networks: The Formation of Common Standards and Visions for Infrastructure Development (pp. 73–94). Routledge.

SRF. (2023, December 12). Österreichischen Bundesbahnen—Preise für Nachtzüge werden stark erhöht – bis zu 200 Prozent. Schweizer Radio und Fernsehen (SRF). https://www.srf.ch/news/international/oesterreichischen-bundesbahnen-preise-fuer-nachtzuege-werden-stark-erhoeht-bis-zu-200-prozent

The 2030 Federal Transport Infrastructure Plan. (n.d.). https://bmdv.bund.de/SharedDocs/EN/publications/2030-federal-transport-infrastructure-plan.pdf?__blob=publicationFile

UIC, DB International, & Sauter-Servaes, T. (2013). UIC-Study Night Trains 2.0, New opportunities by HSR?

Wüpper, T. (2023, June 7). Warum Österreich mit seinen Nachtzügen so erfolgreich ist. Handelsblatt. https://www.handelsblatt.com/unternehmen/handel-konsumgueter/oebb-nightjet-warum-oesterreich-mit-seinen-nachtzuegen-so-erfolgreich-ist/29177326.html

8. Appendices

Appendix 1. Cost calculation

Costs	Name	Unit	Service 1: Low Cost		Service 2: High Speed		Service 3: Day Use		Total Price (€/annum))		
			No.	price/unit	No.	price/unit	No.	price/unit	Price 1	Price 2	Price 3
Initial Investment Cost	Rolling Stock									-	
	Sleeping cars	€/car	0	700.000,00	1	2.860.000,00	1	700.000,00	-	2.860.000,00	700.000,00
	Couchette cars	€/car	4	700.000,00	3	2.860.000,00	3	700.000,00	2.800.000,00	8.580.000,00	2.100.000,00
	Seated and reclining cars	€/car	2	700.000,00	2	2.860.000,00	2	700.000,00	1.400.000,00	5.720.000,00	1.400.000,00
	Locomotives	€/locomotive	1	2.800.000,00	1	2.860.000,00	1	2.800.000,00	2.800.000,00	2.860.000,00	2.800.000,00
Investment cost total (€)									7.000.000,00	20.020.000,00	7.000.000,00
Operating costs	Personnel costs for										
	Train driver	€/hour	8	57,00	7	57,00	8	57,00	172.000,00	136.000,00	344.000,00
	Head of train attendant	€/hour	8	49,00	7	49,00	8	49,00	148.000,00	117.000,00	296.000,00
	Train attendant	€/hour	8	39,00	7	39,00	8	39,00	353.000,00	371.000,00	940.000,00
	Energy costs	€/KWh	9116	0,14	9686	0,14	11737	0,14	480.000,00	510.000,00	1.235.000,00
	Car cleaning costs	€/trips	1	300,00	1	480,00	1	480,00	110.000,00	176.000,00	351.000,00
	Route prices	€/route-km	1097	4,81	905	4,81	1097	4,81	1.926.000,00	1.590.000,00	1.926.000,00
	Station fees	€/stations	1	1.075,84	1	1.075,84	1	1.614,58	393.000,00	393.000,00	590.000,00
	Track access charge	€/km	1097	2,86	905	2,86 or 14,12 on HS	1097	2,86	1.145.000,00	2.807.000,00	3.563.000,00
	Traffic management system	€/annum		20.000,00		30.000,00		40.000,00	20.000,00	30.000,00	40.000,00
	Insurance cost	€/annum		150.500,00		180.500,00		180.500,00	151.000,00	181.000,00	181.000,00
	Inspection cost	€/km	1097	0,84	905	0,84	1097	0,84	1.000,00	1.000,00	2.000,00
	Maintenance cost	€/km	1097	2,76	905	2,76	1097	2,76	1.105.000,00	912.000,00	2.210.000,00
Direct cost total (€/a.)									6.004.000,00	7.224.000,00	11.678.000,00
External cost	Accident costs	€-cent/pkm	296548	0,50	167081	0,50	470484	0,50	148.300,00	84.000,00	236.000,00
	Air pollution costs	€-cent/pkm	296548	0,01	167081	0,01	470484	0,01	3.000,00	2.000,00	5.000,00
	Well-to-tank costs	€-cent/pkm	296548	0,73	167081	0,73	470484	0,73	217.000,00	122.000,00	344.000,00
	Noise costs	€-cent/pkm	296548	0,36	167081	0,36	470484	0,20	107.000,00	61.000,00	95.000,00
External cost total (€/a.)									475.300,00	269.000,00	680.000,00

Appendix 2. Benefit calculation

Benefit	Name	Unit	Service 1: Low Cost		Service 2: High Speed		Service 3: Day Use		Total Price (€)		
			No.	price/unit	No.	price/unit	No.	price/unit	Price 1	Price 2	Price 3
Revenue	Ticket Sales								4.355.000,00	5.993.000,00	8.043.000,00
Direct benefit total (€/a.)									4.355.000,00	5.993.000,00	8.043.000,00
User Benefits	Daytime travel saving	€/h	1,25	26,2	1,5	26,2	1,25	26,2	4.973.000,00	4.074.000,00	3.945.000,00
	Travel comfort	€/annum	2,95		2,63		2,82		1.127.000,00	440.000,00	59.000,00
	Travel time saving for air passenger	€/annum	1,97	26,2	4,47	26,2	1,97	26,2	287.000,00	650.000,00	287.000,00
Indirect benefit total (€/a.)									6.387.000,00	5.164.000,00	4.291.000,00
Social benefits	User groups		0	-	0	-	0	-	-	-	-
Environmental/External ben	carbon emission per passenger km	€-cent/pkm	0	0,29		0,29		0,29	133.000,00	75.000,00	105.000,00
Intangible benefit total (€/a.)									133.000,00	75.000,00	105.000,00
GRAND TOTAL									**10.875.000,00**	**11.232.000,00**	**12.439.000,00**

Appendix 3. Net present value and benefit-cost ratio calculation (thousand €)

	2024			2025			2026			2027		
	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3
Benefits (thousand €)	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436
Costs (thousand €)	13.475	27.506	19.350	6.475	7.486	12.350	6.475	7.486	12.350	6.475	7.486	12.350
Total (thousand €)	- 2.602	- 16.275	- 6.914	4.398	3.745	86	4.398	3.745	86	4.398	3.745	86
Discount rate (3,5%)	*1,000*	*1,000*	*1,000*	*0,966*	*0,966*	*0,966*	*0,934*	*0,935*	*0,935*	*0,902*	*0,902*	*0,902*
Benefits incl. Discount (thousand €)	10.873	11.231	12.436	10.505	10.851	12.016	10.150	10.504	11.632	9.807	10.129	11.217
Costs incl. Discount (thousand €)	13.475	27.506	19.350	6.256	7.232	11.933	6.044	7.001	11.552	5.840	6.752	11.139
Total incl. Discount (thousand €)	- 2.602	- 16.275	- 6.914	4.249	3.619	83	4.106	3.503	80	3.967	3.377	78

	2028			2029			2030			2031		
	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3
Benefits (thousand €)	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436
Costs (thousand €)	6.475	7.486	12.350	6.475	7.486	12.350	6.475	7.486	12.350	6.475	7.486	12.350
Total (thousand €)	4.398	3.745	86	4.398	3.745	86	4.398	3.745	86	4.398	3.745	86
Discount rate (3,5%)	*0,871*	*0,871*	*0,871*	*0,842*	*0,842*	*0,842*	*0,814*	*0,814*	*0,814*	*0,786*	*0,786*	*0,786*
Benefits incl. Discount (thousand €)	9.475	9.787	10.838	9.155	9.456	10.471	8.845	9.136	10.117	8.546	8.827	9.775
Costs incl. Discount (thousand €)	5.642	6.523	10.763	5.452	6.303	10.399	5.267	6.090	10.047	5.089	5.884	9.707
Total incl. Discount (thousand €)	3.833	3.264	75	3.703	3.153	72	3.578	3.046	70	3.457	2.943	68

	2032			2033			2034			2035		
	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3
Benefits (thousand €)	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436
Costs (thousand €)	6.475	7.486	12.350	6.475	7.486	12.350	6.475	7.486	12.350	6.475	7.486	12.350
Total (thousand €)	4.398	3.745	86	4.398	3.745	86	4.398	3.745	86	4.398	3.745	86
Discount rate (3,5%)	*0,759*	*0,759*	*0,759*	*0,734*	*0,734*	*0,734*	*0,709*	*0,709*	*0,709*	*0,685*	*0,685*	*0,685*
Benefits incl. Discount (thousand €)	8.257	8.529	9.444	7.978	8.240	9.125	7.708	7.962	8.816	7.448	7.692	8.518
Costs incl. Discount (thousand €)	4.917	5.685	9.379	4.751	5.493	9.062	4.590	5.307	8.756	4.435	5.127	8.460
Total incl. Discount (thousand €)	3.340	2.844	65	3.227	2.747	63	3.118	2.655	60	3.013	2.565	58

	2036			2037			2038			2039		
	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3
Benefits (thousand €)	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436	10.873	11.231	12.436
Costs (thousand €)	6.475	7.486	12.350	6.475	7.486	12.350	6.475	7.486	12.350	6.475	7.486	12.350
Total (thousand €)	4.398	3.745	86	4.398	3.745	86	4.398	3.745	86	4.398	3.745	86
Discount rate (3,5%)	*0,662*	*0,662*	*0,662*	*0,639*	*0,639*	*0,639*	*0,618*	*0,618*	*0,618*	*0,597*	*0,597*	*0,597*
Benefits incl. Discount (thousand €)	7.196	7.432	8.230	6.952	7.181	7.952	6.717	6.938	7.683	6.490	6.704	7.423
Costs incl. Discount (thousand €)	4.285	4.954	8.173	4.140	4.786	7.897	4.000	4.625	7.630	3.865	4.468	7.372
Total incl. Discount (thousand €)	2.911	2.478	57	2.812	2.395	55	2.717	2.313	53	2.625	2.236	51

	2040			TOTAL			
	Service 1	Service 2	Service 3	Service 1	Service 2	Service 3	
Benefits (thousand €)	10.873	11.231	12.436	184.832	190.912	211.409	
Costs (thousand €)	6.475	7.486	12.350	117.062	147.266	216.949	
Total (thousand €)	4.398	3.745	86	67.766	43.645	- 5.538	
Discount rate (3,5%)	*0,577*	*0,577*	*0,577*				
Benefits incl. Discount (thousand €)	6.271	6.477	7.172	142.365	147.069	162.859	
Costs incl. Discount (thousand €)	3.734	4.317	7.123	91.774	118.044	168.734	
Total incl. Discount (thousand €)	2.537	2.160	49	50.591	29.025	- 5.875	NPV
				1,55	1,25	0,97	BCR

CONTRIBUTIONS

Kiran Chataut

Spreadsheet: Operational cost, indirect benefits: literature review, and its calculations.
Mid and final presentations: 25 %
Report: Methodology- [3.1 (except for sensitivity analysis and travel demand) & 3.3)]
Report layout: 75%

Samuel Juhasz-Aba

Introduction, Project description/proposals, and Methodological framework
Demand estimation
Benefits quantification, (excel) calculations, and literature review.
Mid and final presentations: 25 %, proofreading the report.

Marsa Tyarpratama

Spreadsheet: Formatting, external costs, and sensitivity analysis.
Mid and final presentations: 25 %.
Report: Sensitivity analysis, conclusion, appendixes.

Shyam Twanabasu

Spreadsheet: Demand calculation, benefits, calculations, and literature review.
Mid and final presentations: 25 %.
Report: Demand calculation, Benefits, Results.

Report Writing

Parts	Samuel	Marsa	Kiran	Shyam
1. Introduction	x			
2. Project Description	x			
3. Methodology				
3.1 Introduction and Overview of the CBA Methodology		x	x	
3.2 Demand Estimation for Night Train Services				x
3.3 Costs			x	
3.4 Benefits				x
4. Results	x			x
5. Sensitivity Analysis		x		
6. Conclusion		x		
8. Appendices		x		
Formatting			x	